



**Faculty of Engineering and Technology
Department of Electrical and Computer Engineering**

ENCS5331 Advanced Computer Architecture

Research Project

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2025/2026 Fall Semester

List of The Project Topics

1. Domain Specific Architectures

The team is expected to identify an application domain, e.g., cryptography or AI that needs customized architecture. Then the team should define the custom architecture of this domain, design it, and then simulate it and assess its performance.

2. Accelerators

The team is expected to identify a state-of-the-art compute-intensive and/or data-intensive algorithm, and work on accelerating it through hardware based acceleration or hardware-algorithmic co-acceleration.

3. Hardware Security

This includes hardware techniques (circuit level or architecture level) to protect the secure data in a processor against modern attacks, such as, side channel and fault injection attacks.

4. Microarchitecture

The team is expected to propose a novel technique on the micro-architectural level that leads to a reduction in the latency, power consumption, or area. For example, pre-fetching, branch prediction, instruction execution scheduling, and networks on chip (NoC)

5. Emerging Computing Technologies and Architectures

Any emerging computing technology/architecture the students are interested to investigate, such as, chiplet-based architectures, quantum computing, spin wave computing, neuromorphic computing, computing in memory, and FeFeT transistors.

6. The use of AI in computer architecture, such as, branch prediction and pre-fetching

7. Design space exploration for modern architectures

8. Hardware schedulers in modern AI accelerators

9. Compiler techniques for modern computer architectures

10. Any state of the art topic related to computer architecture from your choice. It requires instructor's approval.

Term Paper Plan

Milestone	Deliverables
Team Building and Topic Selection	- List of up to three team members - Selected Topic (High Level)
Problem Statement	- Clear problem description
Literature Review	- Background section - Related Work section
Proposed Solution	- Clear and detailed description of the adopted solution with block diagrams, flowcharts, pseudocodes, tables, figures, etc. - Justifications of the decisions you made regarding the adopted solution - Methodology
Implementation	- Implementation details - Source code - Demo
Experimental Results	- Experimental setup - Benchmarks - Detailed experimental results - Detailed analysis and discussion of the experimental results - Comparison with others
Final Paper	- Final paper with the following structure: abstract, introduction, background, related work, proposed solution, experimental results, conclusion and future work, acknowledgment, and references
Paper Presentation	

GOOD LUCK